Cryoablation percutanée dans la prise en charge des tumeurs des parties molles : revue pédagogique et technique

Titre original : Percutaneous cryoablation in soft tissue tumor management: an educational review

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Guide pratique pour la cryoablation des masses des parties molles (tumeurs desmoïdes, nodules métastatiques paravertébraux ou du trapèze, schwannomes, malformations vasculaires) en HDJ Radio B.

2. Synthèse Clinique en Français

Résumé exécutif : Revue didactique (*Insights into Imaging*, 2024) détaillant la sélection des patients, la balistique des cryosondes, la synergie thermique et la prévention des complications lors de la cryoablation des tumeurs bénignes et malignes des parties molles (desmoïdes, endométriomes pariétaux, métastases musculaires et sous-cutanées).

Analyse détaillée : Cet article passe en revue l'ensemble des indications de la cryoablation des tissus mous en radiologie interventionnelle oncologique et fonctionnelle. Il détaille l'effet cytotoxique biphasique (formation de cristaux de glace intracellulaires à -40 °C, apoptose périphérique et thrombose microvasculaire), ainsi que la gestion des localisations superficielles ou juxta-nerveuses. Pour les nodules métastatiques musculaires (trapèze, loge paravertébrale) ou les tumeurs desmoïdes, la cryoablation procure un contrôle local durable et une sédation rapide de la douleur par dénervation locale.

3. Protocole Technique & Paramètres Opératoires

Guidage mixte Échographie (insertion des sondes et contrôle Doppler) + Scanner (cartographie 3D de l'iceball) • Gant chaud ou hydrodissection sous-cutanée au sérum physiologique tiède pour protéger le derme • Suivi IRM à 3 mois avec soustraction dynamique.

4. Points Clés à Retenir pour la Pratique

- L'isotherme létal (-20 °C à -40 °C) se situe à environ 3 à 5 mm en dedans de la limite visible du glaçon (0 °C) au scanner.
- Pour les nodules superficiels (trapèze, paroi thoracique), une hydrodissection sous-cutanée continue prévient efficacement la nécrose cutanée.
- Efficacité remarquable sur les localisations oligométastatiques des parties molles (mélanome, cancer du rein, sein, sarcomes).
- Possibilité de réaliser le geste sous anesthésie locale + sédation consciente pour les lésions de petite taille à distance des troncs moteurs.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Background Percutaneous cryoablation (PCA), having shown effectiveness in treating liver, lung, prostate, breast, and kidney tumors, is now gaining attention for the treatment of soft tissue tumors. PCA functions by freezing tissue, which induces ice crystal formation and cell death without damaging collagen structures. Technical considerations include the selection and handling of cryoprobes and cryogenic agents, procedural duration, and choice of image guidance for precision. This review aims to synthesize the mechanisms, applications, and technical aspects of PCA in the treatment of soft tissue tumors. Methods Adhering to PRISMA

2020 guidelines, a review was conducted of studies published prior to March 2024 that investigated PCA of soft tissue tumors. The review focused on technical and procedural aspects of cryoablation, cryobiological principles, cellular and tissue responses to extreme cold, intra- and post-procedure physiological mechanisms during and post-procedure, and main clinical applications. Results PCA is efficient in treating soft tissue tumors, including desmoid tumors, vascular malformations, and abdominal wall endometriosis. Several cryobiological mechanisms are involved, notably ice crystal formation, cellular dehydration, osmotic effects, and the inflammatory response, all of which contribute to its efficacy. Key technical aspects include the choice of cryoprobes, cryogenic agents (argon gas or liquid nitrogen), and the duration and control of freezing/thawing cycles. PCA also frequently outperformed traditional treatments like surgery and radiotherapy in terms of pain reduction, tumor size reduction, and patient outcomes. Moreover, its nerve sideration properties make it effective under local anesthesia. Conclusion Demonstrating substantial pain reduction, tumor size decrease, and high technical success rates, PCA offers a promising and minimally invasive alternative for soft tissue tumor treatment. Critical relevance statement Percutaneous cryoablation provides a minimally invasive, precise alternative for soft tissue tumor management, advancing clinical radiology by offering effective treatment with reduced patient risk and enhanced outcomes through image-guided procedures. Key points Percutaneous cryoablation (PCA) offers a promising, minimally invasive alternative for managing soft tissue tumors. PCA employs image-guided techniques to accurately target and treat tumors, ensuring high precision and control. PCA preserves structures like collagen, reduces pain, decreases tumor size, and generally enhances patient outcomes.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Introduction

Soft tissue tumors, with an incidence of about 300 cases per 100,000 people [1], exhibit a wide range of complex and diverse histological presentations, including benign lesions such as abdominal wall endometriosis (AWE) and vascular malformations, desmoid tumors, and malignant lesions such as sarcomas [2]. Benign lesions are more common, while malignancies account for less than 1% [3, 4].

Percutaneous image-guided ablation is increasingly used as a substitute for surgical procedures in various cases, especially in patients for whom surgery poses risks or is deemed too complicated [5]. Percutaneous cryoablation (PCA), which involves the image-guided insertion of cryoprobes into tissue, is mainly used for tumors of the liver, lung, prostate, breast, and kidney. Recently, interest has grown in the effectiveness of PCA for the treatment of soft tissue tumors [2, 6]. Unlike radiofrequency ablation (RFA), high-intensity focused ultrasound, or microwave ablation, PCA does not compromise fibrillar structures like collagen. Collagen fibers consist of long, triple-helical protein strands that are less susceptible than cellular water content to the formation of ice crystals during freezing. Furthermore, PCA provides precise control over the ablation zone, and its real-time imaging guidance ensures accurate targeting of tumors, reducing the risk of damage to surrounding healthy tissues. Cold exposure can also slow nerve conduction, reducing nerve activity and dampening pain signals, which produces an anesthetic effect. This improves patient tolerance to the procedure [6–9]. These advantages make PCA a preferred treatment compared to other ablation techniques.

This educational review aimed to provide a comprehensive synthesis of PCA mechanisms for soft tissue tumors, offering an overview of PCA’s primary applications while considering the distinct anatomical and tumoral constitution of soft tissue as well as PCA’s technical and practical considerations.

Methods

An analysis of studies published prior to March 2024 that focused on PCA of soft tissue tumor was conducted in accordance with the PRISMA 2020 guidelines, with registration in NIHR-PROSPERO (Fig. 1).Fig. 1Research methodology

This study aimed to assess the technical and procedural aspects of PCA in soft tissue, emphasizing the cryobiological principles underlying its effectiveness. The study included a comprehensive analysis of cellular

and tissue responses to extreme cold temperatures, the physiological mechanisms activated intra- and post-procedure, and a review of main clinical applications.

This study detailed the cryoprobes utilized, the stepwise procedural workflow, the types of cryogenic agents used, and the role of imaging guidance in enhancing precision and safety. It also reported the efficacy and adverse effects (AE) of PCA in treating main soft tissue tumors.

A search of the PubMed (MEDLINE) database was conducted on March 1, 2024, to identify relevant literature using a predefined strategy. This platform was chosen for its accessible and high-quality peer-reviewed articles. Table 1 details the search terms. Two reviewers independently screened each article, including references from pertinent systematic reviews and meta-analyses.Table 1Search termsCryoablation OR CryotherapyANDTechnique OR Cryobiology OR Mechanisms OR Technology OR PhysiologyANDSoft tissue tumor OR Desmoid tumor OR Desmoid fibromatosis OR Vascular malformations OR Abdominal wall endometriosisNOTLaparoscopic/surgery

Criteria for inclusion comprised studies focusing on PCA mechanisms in soft tissue. Exclusion criteria were studies with only abstracts available, non-English language publications, non-human studies, editorial-style reviews, abstracts and posters, conference papers, and case reports. Based on these criteria, two authors assessed full-text articles. In addition to the parameters above, this study also involved a thorough review of PCA's primary clinical applications, focusing mainly on desmoid tumors, vascular malformations, and AWE. This review aimed to understand PCA's practical implications and outcomes in these specific conditions, thereby examining its effectiveness, patient outcomes, and any noted complications or challenges.

A specifically designed form was used to extract essential information from selected studies. Two radiologists conducted this extraction process independently, with discrepancies resolved through a consensus involving a third senior radiologist. The Quality In Prognosis Studies tool was utilized to control the risk of bias assessment.

Results

PCA freezes tissue in order to intracellularly induce ice crystals that cause cell death. Unlike other percutaneous ablation techniques (radiofrequency ablation, high-intensity focused ultrasound, laser, and microwave ablation), PCA preserves fibrillar structures like collagen, without causing damage. Collagen fiber is made up of extended, triple-helix protein strands that are more resistant to ice crystal formation during freezing than cellular water content. Exposure to cold can decrease nerve conduction velocity, which leads to attenuation of pain signals. The resulting anesthetic effect [6, 8, 9] enhances patient comfort during the procedure and enables the execution of PCA using local anesthesia alone. As transient neuropraxia is observed below +5 °C and definitive injuries are observed below –20 °C, the duration of the nerve block depends on various factors, including the temperature reached. Thus, understanding PCA's effects is crucial for the comprehension of the mechanisms through which freezing temperatures impact cellular and tissue structures in soft tissue. These mechanisms contribute to PCA's therapeutic outcomes [9, 10]. PCA leads to the following five common effects [6, 8, 9].

As PCA temperatures drop below –40 °C, water within the cells begins to freeze and form ice crystals. Ice crystal formation can cause physical damage to cellular structures, including the cell membrane, organelles, and cytoskeleton. The sharp edges of ice crystals can pierce cell membranes, disrupting the integrity of the cells. Changes in membrane permeability can disrupt the balance of ions and molecules within the cell. There is a mismatch between the –40 °C isotherm and the volume of the ice ball, which explains why the cycle must be considered, as well as the extension of margins beyond the boundaries of the lesion [6, 8, 9].

Osmotic effects play a significant role in the cellular response to the freezing and thawing process that occurs at temperatures below 0 °C. Osmosis refers to the movement of water across a semipermeable membrane from an area of low solute concentration to an area of high solute concentration. As tissue is exposed to freezing temperatures, water within the cells begins to freeze and form ice crystals. This process results in relative water extraction from the intracellular space, meaning that although the water remains within the cell, it transforms into ice. This transformation effectively removes the water, causing an increase in the concentration

of solutes in the unfrozen water and leading to cellular dehydration. Cellular dehydration can contribute to both cell shrinkage and alteration of intracellular structures. The augmentation of the concentration of solutes, creating an osmotic gradient between the intracellular and extracellular spaces, can affect the structural integrity of cell membranes and organelles, leading to a compensatory influx of water into the cell. Multiple cycles will optimize this effect throughout the ice ball, leading to cell death within the ice ball's whole volume [6, 8, 9].

Unplanned cell death (necrosis) occurs, as does the induction of programmed cell death (apoptosis) by compromising the structural integrity and function of cellular organelles, such as mitochondria, endoplasmic reticulum, and nuclei. Mitochondrial damage particularly impacts cellular energy production and contributes to cell death. Apoptosis is triggered by various cellular signals, including those related to osmotic stress and damage to cellular structures [6, 8, 9].

The release of various signaling molecules and the recruitment of immune cells to the injury site result in delayed inflammation after ice ball melts may last up to 6 months. This inflammatory response contributes to the removal of damaged cells and tissue as part of the healing process. It also contributes to the abscopal effect [11] and may be promoted by adjunction of immune therapies [12].

Osmotic effects contribute to vasoconstriction, or the reduction of blood flow to treated tissue. Ice also occludes the vessels with a diameter less than 3 mm, which explains why the intensity of this vascular effect may be related to freeze duration. Both contribute to tissue damage by creating an ischemic environment and then necrosis. Reduced blood flow may also help to minimize intraprocedural bleeding, but does not affect post-procedural bleeding when ice melts. Bleeding risk can be mitigated by avoiding the puncture of large vessels, or by using active thawing at the end of the procedure to avoid cracks in the ice. A slow passive thaw should be prioritized, and the needle should be removed without any active thawing [6, 8, 9].

Figure 2 summarizes these five common effects of PCA.Fig. 2Illustration of the effects of cryoablation

Illustration of the effects of cryoablation

Discussion

PCA distinguishes itself from other ablation methods by its ability to preserve collagen structures while inducing cell death through ice crystal formation. This technique's effectiveness is enhanced by cryoprobes that enable precise control over ice formation and accurate targeting of lesions. The choice of cryogenic agent is pivotal, with agents like argon gas or liquid nitrogen presenting unique properties that contribute to the freezing process.

Procedural success hinges on real-time imaging technologies like computed tomography, MRI, or ultrasonography, which guide the placement of cryoprobes and monitor ice ball formation. Temperature-sensitive techniques and protective measures permit the preservation of surrounding healthy tissues. Furthermore, PCA's anesthetic effects—notably its ability to dampen nerve activity and pain signals—enhance patient tolerance. The flexibility in probe selection and ability to optimize the freeze/thaw cycle allow for a tailored approach to each patient's needs.

PCA's combination of precision and control makes it a promising tool for the treatment of soft tissue tumors. Bodard et al's [2] systematic review of 393 PCA procedures for desmoid tumor treatment revealed significant pain and tumor size reductions, with an overall objective response rate of 80%. PCA's performance was equal to or superior to traditional treatments like surgery, radiotherapy, and chemotherapy. The same review reported a 100% success rate for PCA treatment of vascular malformations, with substantial pain reduction and improvements in quality-of-life indicators. The review also included an analysis of 103 PCA sessions of AWE treatment, revealing a high rate of pain reduction and efficacy, as well as a recurrence rate significantly lower than surgery. While surgical treatments for AWE present risks like severe AEs and aesthetic concerns, PCA could maintain a profile of minimal AEs. PCA has also expanded its scope to include other soft tissue tumors, such as sarcomas [9], retroperitoneal metastases, pelvic sidewall disease, and gynecologic lesions, without the risks linked to surgery or systemic therapies [30]. While interest in PCA use for a range of soft tissue tumors is

increasing, the current body of research comprises mainly smaller single studies. A more comprehensive scope of research is required to fully evaluate the efficacy and safety of PCA in these diverse applications. Further, consideration of recent economic factors, such as the increased cost of helium and argon—both critical components of cryoablation procedures—is essential. Similarly, rising expenses may have significant implications for the overall accessibility and affordability of cryoablation therapies in diverse healthcare settings.

In conclusion, PCA’s ability to target lesions with minimal damage to surrounding tissues while offering advantageous anesthetic properties significantly enhances treatment outcomes. Indeed, PCA demonstrates a good success rate and patient tolerance in soft tumor treatment. Nevertheless, and despite its promising results in current studies, there remains a need for broader research to fully ascertain the efficacy and safety of PCA across a wider range of applications.